CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

8.3 Interference

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

Two fringes were created on a screen in a double-slit experiment; the distances were too small to measure.

Which of the following would increase the distance between the fringes?

A increasing the frequency of the light source

B increasing the distance between the light source and the slits

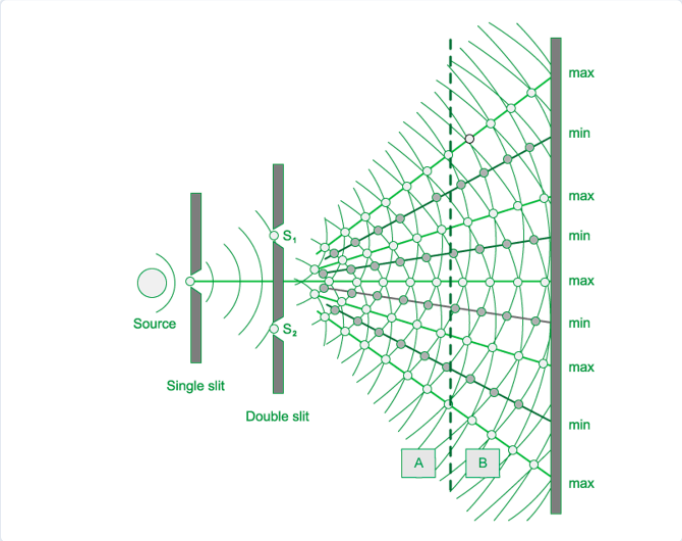
C increasing the distance between the slits and the screen

D increasing the distance between the slits

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- In the diagram below you can see that screen A is closer than screen B



- Fringe separation $\Delta x = \frac{\lambda D}{a}$ where a is the slit separation, λ is the wavelength of light and D the distance from the slits to the screen

A is incorrect because increasing the frequency will decrease the wavelength, as the wavelength decreases this will reduce the diffraction

B is incorrect because increasing the distance between the light source and the slits will change the intensity of the light $I \propto \frac{1}{a^2}$

D is incorrect because increasing the distance between the slits would reduce the number of slits in the diffraction grating and reduce the intensity of the maxima lines

Question 2 Easy [1 mark]

A student investigates the interference of light using two identical green LEDs.

Which statement would explain why the investigation will not work?

A the light waves from the sources have a range of wavelengths

B the light waves from the sources are not monochromatic

C the light waves from the sources are not coherent

D the light waves from the sources do not have the same amplitude

Mark Scheme & Explanation:

*The correct answer is **C** because:*

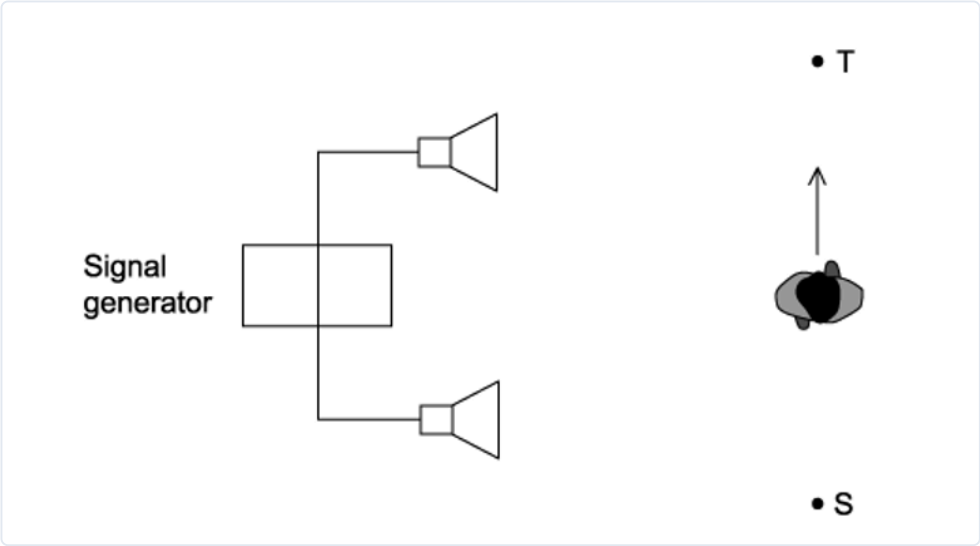
- In order to observe interference in light waves, the following conditions must be met:
  - The source must be coherent – they must have the same phase
  - The source must be monochromatic – the light must be of a single wavelength
- Interference occurs when light waves are coherent (they have a constant phase difference and the same frequency)
- If the light waves are not coherent, the experiment will not work

A & **B** are incorrect because the LED's are identical, they are monochromatic and have the same wavelength

D is incorrect because amplitude is not a condition for interference

Question 3 Easy [1 mark]

Two loudspeakers are connected to a signal generator, as shown in the diagram.



As the student walks between S to T she notices that the loudness of the sound increases then decreases repeatedly.

Why does the loudness change?

A polarisation of the sound waves

B reflection of the sound waves

C diffraction of the sound waves

D interference of the sound waves

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- When two or more sound waves from different sources are emitted at the same time, they interact with each other
    - This is called interference
- If the compressions and rarefactions of the two waves line up, they constructively interfere, creating a wave with a higher intensity or loudness
- As the student walks from point S to T, the sound will increase in amplitude due to the constructive interference

A is incorrect because sound waves are longitudinal waves, and longitudinal waves can't be polarised

B is incorrect because reflection of sound waves is an echo and will not increase the amplitude of the sound

C is incorrect because diffraction happens when a wave meets an object or slit

Question 4 Easy [1 mark]

Fig 1.1 shows two waves

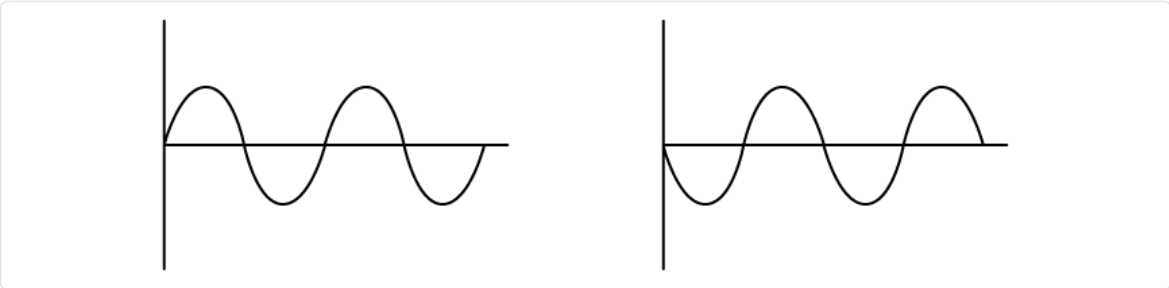


Fig 1.1

Which of the following statements is true?

A The waves are coherent

B The waves are incoherent

C The waves are in phase

D When added together, the waves would cause constructive interference

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The waves are coherent as they have the same frequency and a constant phase difference

Option **B** is incorrect because

- The waves are coherent

Option **C** is incorrect because

- When one wave has a peak, the other wave has a trough
    - Therefore the waves are 180° out of phase

Option **D** is incorrect because

- When added together, the waves would cause destructive interference
    - This is because the waves are 180° out of phase
    - The waves will cancel each other out

Question 5 Easy [1 mark]

Monochromatic light of wavelength λ is incident upon a double slit and forms an interference pattern on a screen.

What is a possible path difference for a minima that appears on the screen?

A 0

B $\frac{3}{2}\lambda$

C $3 \quad \lambda$

D $5 \quad \lambda$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Minimas occur when there is destructive interference
- Destructive interference happens when the path difference is a whole number of wavelengths plus half a wavelength
  - Path difference = $n \quad + \quad \frac{1}{2}\lambda$
  - Path difference = $1 \quad + \quad \frac{1}{2} \quad \lambda \quad = \quad \frac{3}{2}\lambda$
- This is option **B**

Option **A** is incorrect because

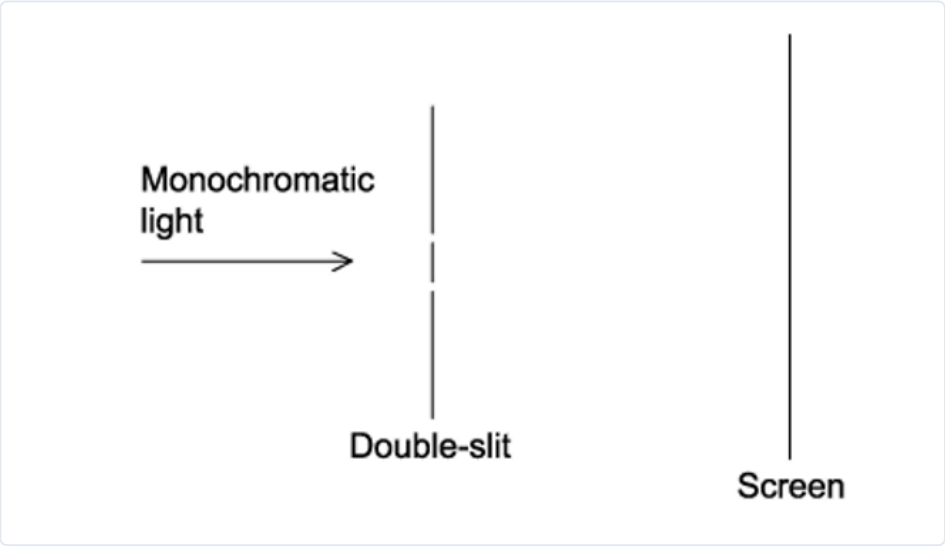
- A path difference of zero is the central maxima on the screen

Option **C** and **D** is incorrect because

- Maximas occur when there is constructive interference
- Constructive interference happens when the path difference is a whole number of wavelengths
  - Path difference = $n\lambda$
- 3λ and 5λ are both whole number of wavelengths
  - Maximas will occur at these path differences

Question 6 Medium [1 mark]

The effect of a double slit on monochromatic light is observed by set up in the diagram



Interference fringes are seen on the screen.

Which of the following changes would increase the distance between the adjacent fringes?

A decrease the width of each slit

B decrease the distance between the two slits

C use light of a higher frequency

D move the screen closer to the double-slit

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The fringe separation equation is given by
  - $x = \frac{\lambda D}{a}$
- Consider option B:
  - If we decrease the distance a between the two slits, this will increase the fringe separation x

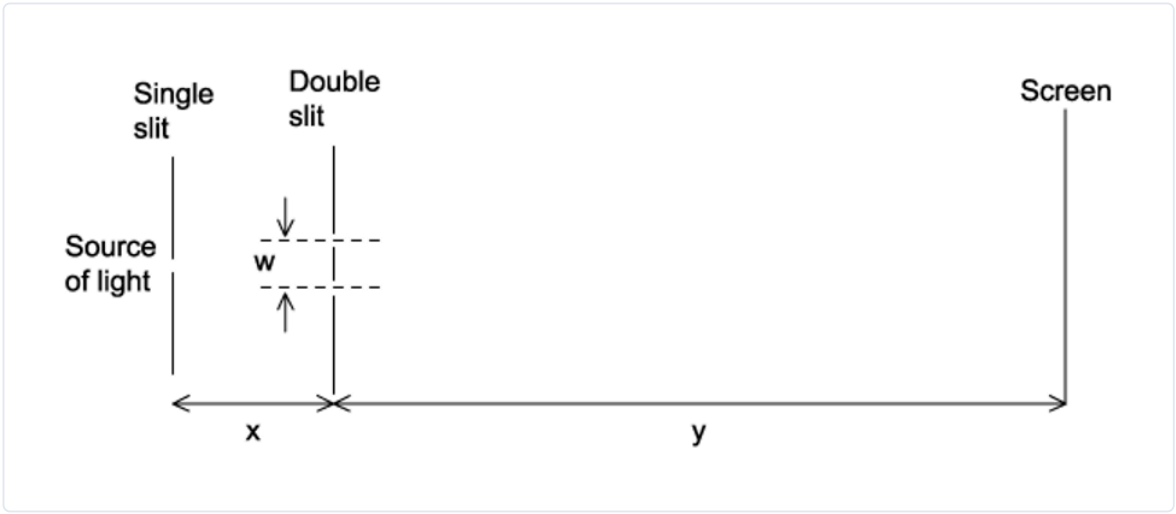
A is incorrect because the width of each slit is not a quantity in the fringe spacing equation

C is incorrect because using light of a higher frequency will result in a smaller wavelength λ, and hence cause a decrease in fringe spacing x

D is incorrect because moving the screen closer to the double-slit will decrease the value of *D* and hence the value of fringe spacing *x* will decrease

Question 7 Medium [1 mark]

The equipment was set up as in the diagram to investigate the interference of a double slit.



Which of the following changes would increase the fringe spacing?

A decreasing the wavelength of the light

B decreasing distance *y*

C decreasing distance *x*

D decreasing the distance *w*

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The fringe separation equation is given by
  - Fringe separation $= \frac{\lambda x}{w}$
- Consider option D:
  - Decreasing the distance *w*, or slit separation, causes the fringe separation to increase

A is incorrect because decreasing the wavelength λ of the light will cause the fringe separation to decrease

B is incorrect because decreasing the distance *y* between the slits and the screen will cause the fringe separation to decrease

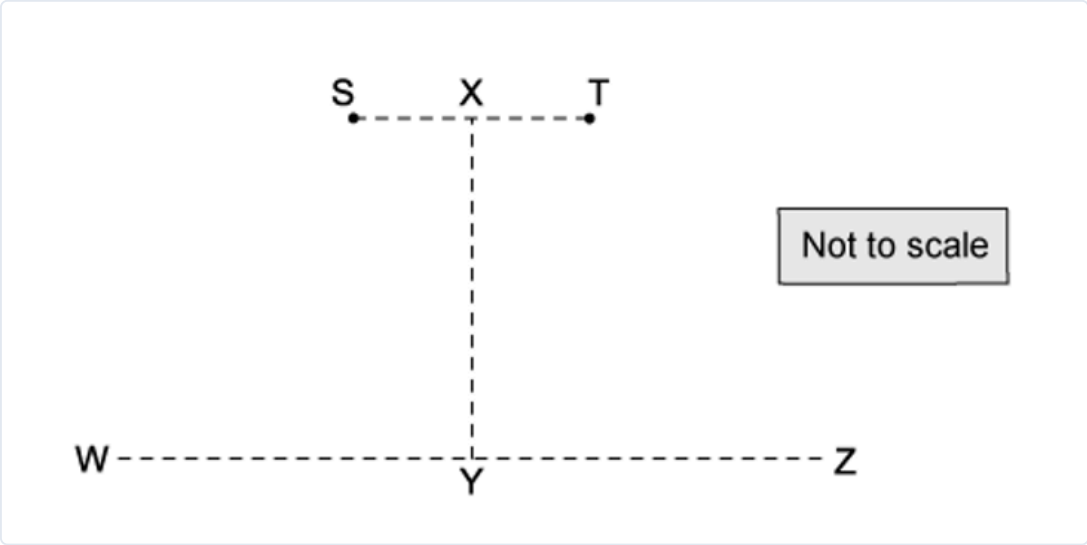
C is incorrect because the distance *x* is not a quantity in the fringe spacing equation

> Don't let the notation used in this question confuse you. It different to that used in your course specification and the revision notes.

Question 8 Medium [1 mark]

A student is investigating the interference of waves. They set up coherent waves produced at points S and T that travel outwards in all directions.

The line XY is halfway between S and T and perpendicular to the joining line S and T. The distance XY is much greater than the distance ST.



Which line or lines would an interference pattern be seen on?

A WZ only

B XY only

C Both XY and WZ

D Neither XY or WZ

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- An interference pattern consists of a series of alternating constructive interference bright spots and destructive interference dark spots
    - An interference pattern will not be observed along a line where only constructive or destructive interference occurs
- The waves are produced at S and T – the phase difference between the waves from S and T is constant along line XY since the waves are coherent and XY is midway from S and T

- Therefore, the path difference along XY is always zero — that means, only constructive interference occurs along this line (for constructive interference, path difference where n = 0, 1, 2, 3 etc.)
- So, there can be no interference pattern along XY
    - Therefore, an interference pattern will only observed along line WZ

Question 9 Medium [1 mark]

Certain circumstances will produce visible interference fringes from two light sources.

Which of the following conditions will enable visible interference fringes to be seen?

A using one light which is polarised at right angles to light from the other source

B using incoherent sources

C using a blue light source

D using sources from which the light does not overlap

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- In order to observe interference in light waves, the following conditions must be met:
    - The source must be coherent – they must have the same phase
    - The source must be monochromatic – the light must be of a single wavelength
- Using blue light of a single wavelength satisfies the second condition

A is incorrect because light will only interfere with a light of the same polarisation

B is incorrect because light will only interfere if both sources are coherent

D is incorrect becuse the light sources must overlap in order for interference occur

Question 10 Medium [1 mark]

Monochromatic light produces interference fringes with a separation of x on a screen a distance D from a pair of slits with separation a.

The student doubles a and D.

What would be the new fringe separation?

A $2x$

B $4x$

C x

D $\frac{x}{2}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The fringe separation equation is given by
  - $x = \frac{\lambda D}{a}$
- Where:
  - x is the fringe separation
  - λ is the wavelength
  - $2D$ is the distance between the slits and the screen
  - $2a$ is the slit separation
- So, the new fringe spacing is equal to:
  - $x = \frac{\lambda 2D}{2a} = \frac{\lambda D}{a}$
- Therefore, the value of the fringe spacing x does not change

Question 11 Medium [1 mark]

Red light with a wavelength of 650 nm is used in a Young's double-slit experiment. The distance between the double slit and the screen is 2.5 m and the separation of the slits is 0.80 mm.

What would be the fringe separation on the screen?

A 0.13 mm

B 0.21 mm

C 1.3 mm

D 2.0 mm

Mark Scheme & Explanation:

The correct answer is D

- The fringe separation equation is given by

$$x = \frac{\lambda D}{a}$$

- Where:
  - x is the fringe separation
  - a is the slit separation = 0.80 mm
  - λ is the wavelength = 650 nm
  - D is the distance = 2.5 m

• Substituting the values:

$$x = \frac{650 \times 10^{-9} 2.5}{0.80 \times 10^{-3}} = 2.0 \times 10^{-3} \text{ m} = 2.0 \text{ mm}$$

- Therefore, the fringe separation on the screen is 2.0 mm

Question 12 Hard [1 mark]

Yellow light with a wavelength of 600 nm is used in an experiment to show interference fringes produced by double-slit interference. The fringe separation is 4.0 mm and the separation of the slits is 0.60 mm.

What would be the distance between the double slit and the screen?

A 4.0 m

B 2.5 m

C 0.4 m

D 0.25 m

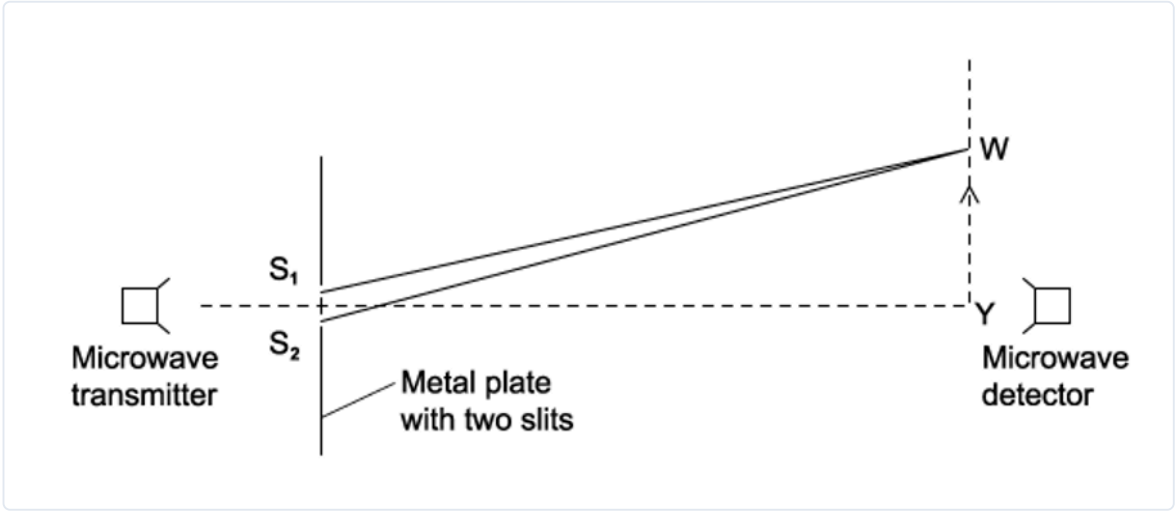
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The fringe separation equation is given by
  - $x = \frac{\lambda D}{a}$
- Where:
  - x is the fringe separation = 4.0 mm
  - a is the slit separation = 0.60 mm
  - λ is the wavelength = 600 nm
- So, rearranging for distance D:
  - $D = \frac{xa}{\lambda} = \frac{4.0 \times 10^{-3}\, 0.60 \times 10^{-3}}{600 \times 10^{-9}} = 4.0 \text{ m}$
- Therefore, the distance between the double slit and the screen is 4.0 m

Question 13 Hard [1 mark]

An experiment was set up to investigate two-source interference. Microwaves of wavelength λ are passed through S_1 and S_2.



The detector is moved in the direction of the arrow from Y to W. The signal detected is decreased to zero at point W and then starts to increase again as it continues past W.

Which of the following would be the equation to determine the position of point W?

A $S_2W - S_1W = \frac{\lambda}{2}$

B $S_2W - S_1W = \lambda$

C $YW = \frac{\lambda}{2}$

D $YW = \lambda$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A minima is found at W since the signal keeps on decreasing until it reaches W, so, at W, the signal has the lowest amplitude
    - Therefore, W is the first minima: this is where destructive interference takes place
- S_2W and S_1W are the distances between the slits and the screen, hence the path difference is equal to $S_2W - S_1W$
- The condition for destructive interference is a path difference $\frac{n\lambda}{2}$
- For the first minima (n = 1):

    - Path difference $= \frac{\lambda}{2}$
    - S_2W - $S_1W = \frac{\lambda}{2}$
- So, the distance $\frac{\lambda}{2}$ is equal to the path difference between S_2W and S_1W which is S_2W - S_1W

B is incorrect because this would be at the first maxima, where path difference = $n\lambda$

C & D are incorrect because

- Fringe separation is equal to $x = \frac{\lambda D}{a}$
- The distance between the central maxima (Y) and the first minima (at W) is half the fringe separation $YW = \frac{x}{2}$
- So, YW is NOT equal to $\frac{\lambda}{2}$ or λ

Question 14 Hard [1 mark]

Yellow light with a wavelength of 600 nm is used in an experiment to show interference fringes produced by double-slit interference. The fringe separation is 4.0 mm on the screen.

What would be the fringe spacing if the wavelength is changed to 400 nm and the separation of the slits is doubled?

A 1.3 mm

B 3.0 mm

C 5.3 mm

D 12.0 mm

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The fringe separation equation is given by
  - $x = \frac{\lambda D}{a}$
- Where:
  - x_1 is the fringe separation = 4.0 mm, which changes to x_2
  - λ_1 is the wavelength = 600 nm, which changes to λ_2 = 400 nm
  - a is the slit separation, which changes to $2a$
  - D is the distance between the double slit and the screen, which does not change
- So, since distance D does not change, we should rearrange for D:
  - $D = \frac{xa}{\lambda}$
  - $D = \frac{x_1 a_1}{\lambda_1} = \frac{x_2 a_2}{\lambda_1}$
- Substitute $a_2 = 2a_1$ into the equation
  - $\frac{x_1 a_1}{\lambda_1} = \frac{x_2 2a_1}{\lambda_2}$
  - $\frac{x_1}{\lambda_1} = \frac{2x_2}{\lambda_2}$
- Rearrange for the new fringe spacing x_2:
  - $x_2 = x_1 \frac{\lambda_2}{\lambda_1} \frac{1}{2}$

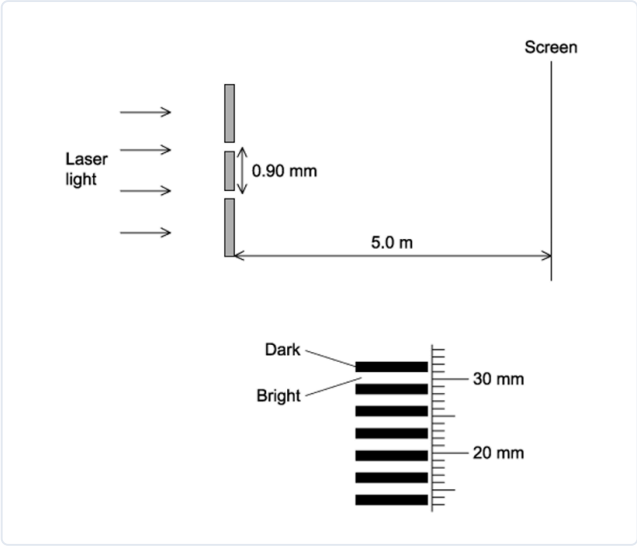
- $x_2 = 4.0 \times \frac{400}{600} \times \frac{1}{2} = 4.0 \times \frac{1}{3}$
- $x_2 = 1.33 \text{ mm}$

- Therefore, the new fringe spacing is 1.3 mm

You can also solve this question by figuring out the ratio required to move from x_1 to x_2. The wavelength changes by a multiple of $\frac{2}{3}$.

Question 15 Hard [1 mark]

A student carried out an experiment on Young's slits using a laser light. There is also a section of the pattern produced with a ruler placed next to it.



What is the wavelength of the light?

A 540 nm

B 3.4 μm

C 480 nm

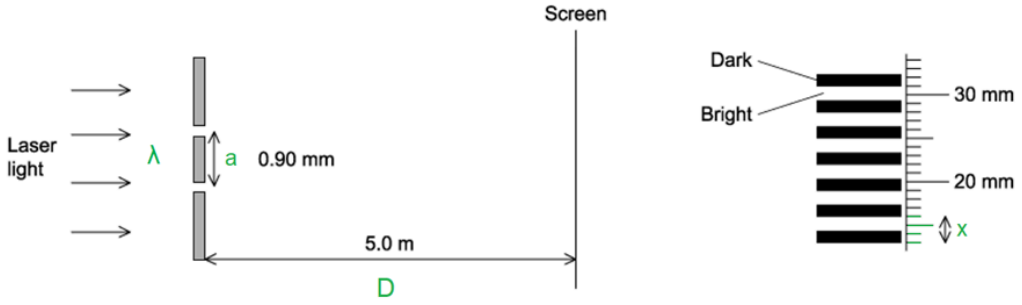
D 3.2 μm

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The fringe separation equation is given by
  - $x = \frac{\lambda D}{a}$
- Where:
  - x is the fringe separation = 3.0 mm
  - a is the slit separation = 0.90 mm
  - D is the distance between the double slit and the screen = 5.0 m
  - λ is the wavelength of the light

- Rearranging for wavelength λ
  - $\lambda = \frac{xa}{D}$
  - $\lambda = \frac{3.0 \times 10^{-3} 0.90 \times 10^{-3}}{5.0} = 5.4 \times 10^{-7}$ m
- Therefore, the wavelength of the light is equal to 540 nm



Question 16 Hard [1 mark]

A beam of monochromatic light with wavelength λ is shone through a double slit at distance D away from a screen. An interference pattern is produced where the distance between the fringes is x.

What would the distance between the double slit and the screen need to be for the fringe distance to remain x when a light beam of wavelength $\frac{3}{4}\lambda$ is used instead?

A $\frac{1}{4}D$

B $\frac{1}{3}D$

C $\frac{3}{4}D$

D $\frac{4}{3}D$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The known quantities are
  - The distance between the slit and the screen when
- The double-slit equation is
  - $\lambda = \frac{ax}{D}$
- The distance between the slits a and fringes x will remain the same
  - $\lambda \propto \frac{1}{D}$
- When the wavelength is equal to λ
  - $D \propto \frac{1}{\lambda}$
- When the wavelength is equal to $\frac{3}{4}\lambda$
  - $\frac{3}{4}\lambda \quad \propto \frac{1}{D_2}$
  - $D_2 \quad \propto \quad \frac{4}{3\lambda}$
- The ratio of D_2 and D is
  - $\frac{D_2}{D} = \frac{4}{3\lambda} \times \frac{\lambda}{1} = \frac{4}{3}$
  - $D_2 = \frac{4}{3}D$
- This is option **D**